THE OPENAI LOWER BOUND ON $R_k(3)$

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Abstract. We give a gentle exposition of a construction, recently discovered by OpenAI, which shows that the k-colour Ramsey number of the triangle satisfies

$$R_k(3) \geqslant \left(\frac{ck^{1/3}}{\log k} \right)^k$$

for some constant $c > 0$ and every $k \geqslant 2$.

1. Introduction

The k-colour Ramsey number of the triangle, $R_k(3)$, is the smallest $n \in \mathbb{N}$ such that every k-colouring of the edges of K_n contains a monochromatic copy of K_3. It is easy to show, using the classical algorithm of Erdős and Szekeres [3], that

$$R_k(3) = O(k!),$$

but until a few weeks ago, the best known lower bound was of the form

$$R_k(3) \geqslant 380^{k/5} - O(1),$$

which follows from bounds on the Schur numbers proved in [1, Corollary 2.9]. The new construction (see [4, Chapter 9]) gives the following huge improvement over this bound.

Theorem 1.1 (OpenAI, 2026)**.** *There is a constant $c > 0$ such that*

$$R_k(3) \geqslant \left(\frac{ck^{1/3}}{\log k} \right)^k$$

for every $k \geqslant 2$.

The proof is via a simple and elegant random inductive construction. I will begin by describing the first step of this construction, which illustrates most of the key ideas in a simpler setting where they are somewhat easier to understand, and then explain how the construction can be iterated to give the claimed bound.

2. The first step of the construction

The starting point of the construction is the simple (and classical) fact that if $n = 2^k$, then the edges of K_n can be covered by k bipartite graphs. Note that this implies that

$$R_k(3) \geqslant 2^k,$$

by thinking of each bipartite graph as a colour class. In this section we will show how to extend this construction to give a colouring which shows that

$$R_{2k}(3) \geqslant 2^{3k-o(k)},$$

and in which moreover every colour class is tripartite. In the process, we will explain most of the new ideas that we will need in order to prove Theorem 1.1.

Theorem 2.1. *For some $n = 2^{3k-o(k)}$, there exists a colouring of the edges of K_n with $2k$ colours such that each colour forms a triangle-free tripartite graph.*

The idea is to take $2^{2k-o(k)}$ disjoint copies of any k-colouring φ of K_{2^k} with bipartite graphs, but instead of using the same set of k colours for each (as is usually done in such inductive constructions), we use a different set of k colours for each. We moreover choose the sets of colours so that they are pairwise far apart, using the following lemma.

Lemma 2.2. *If $d = o(k)$, then there exists a family $\mathcal{F}$ of subsets of $[2k]$ of size k with*

$$|\mathcal{F}| = 2^{2k-o(k)} \qquad \text{and} \qquad |S \triangle T| \geqslant 2d \tag{1}$$

for every $S, T \in \mathcal{F}$ with $S \neq T$.

Proof. Choose the sets in $\mathcal{F}$ greedily, one at a time, from the k-sets at distance at least $2d$ from all earlier elements of $\mathcal{F}$. We can continue doing so as long as

$$|\mathcal{F}| \sum_{i=0}^{d-1} \binom{k}{i}^2 < \binom{2k}{k},$$

and since $d = o(k)$ it follows that we can continue until $|\mathcal{F}| = 2^{2k-o(k)}$, as claimed. □

We now need to define the colours of the edges between the $|\mathcal{F}|$ copies of φ. To do so, fix a pair $S, T \in \mathcal{F}$, let X and Y be the corresponding copies of φ, and choose sets

$$A \subset S \setminus T \qquad \text{and} \qquad B \subset T \setminus S, \tag{2}$$

with $|A| = |B| = d$, noting that this is possible because $|A \triangle B| \geqslant 2d$, by (1). In other words, the colours in A are used in X but not Y, and the colours in B are used in Y but not X. We will only use colours in the set $A \cup B$ for the edges between X and Y.

We now make a simple but key observation.

Observation 2.3. *For each colour $c \in [2k]$, the edges of colour c with endpoints in different copies of φ form a bipartite graph.*

Proof. By (2), each edge of colour c with endpoints in different copies of φ has one vertex in the set of copies that use the colour c, and one in the set of copies that don't. □

Observation 2.3 has two important consequences:

1. There are no monochromatic triangles with vertices in three different copies of φ.
2. The edges of each colour form a tripartite graph, since the copies of φ that use c are bipartite in colour c, and the remaining copies are an independent set.

It will therefore suffice to define a colouring between X and Y, using the colours $A \cup B$, that does not create any monochromatic triangles in $X \cup Y$.

To do so, we will need a little notation: for each vertex $u \in X \cup Y$ and colour $c \in A \cup B$, let

$$N_c(u) = \big\{ v \in X \cup Y : uv \text{ is an edge of colour } c \big\},$$

and let $U_c(0)$ and $U_c(1)$ be the parts of the bipartite graph formed by the edges of colour c in X or Y. To avoid monochromatic triangles, it will suffice to ensure that

(a) For each colour $a \in A$ and each vertex $y \in Y$, either

$$N_a(y) \cap X \subset U_a(0) \qquad \text{or} \qquad N_a(y) \cap X \subset U_a(1).$$

(b) For each colour $b \in B$ and each vertex $x \in X$,

$$N_b(x) \cap Y \subset U_b(0) \qquad \text{or} \qquad N_b(x) \cap Y \subset U_b(1).$$

To define such a colouring, let us map each element $x \in X$ to the hypercube $\{0,1\}^A$ by encoding, for each colour $a \in A$, whether $x \in U_a(0)$ or $x \in U_a(1)$, and similarly map Y to $\{0,1\}^B$. We will define a colouring between $\{0,1\}^A$ and $\{0,1\}^B$ with the claimed property, and then blow it up to give a colouring between X and Y.

To define the colouring between $\{0,1\}^A$ and $\{0,1\}^B$, we will choose two functions

$$f \colon \{0,1\}^A \to \{0,1\}^B \qquad \text{and} \qquad g \colon \{0,1\}^B \to \{0,1\}^A$$

that encode the set of *permitted* colours for each edge xy with $x \in \{0,1\}^A$ and $y \in \{0,1\}^B$. To be precise, we say that the colour i is permitted for xy if either

$$f(x)_i = y_i \;\text{ and }\; i \in B \qquad \text{or} \qquad g(y)_i = x_i \;\text{ and }\; i \in A.$$

Observe that if each edge uses a permitted colour, then the resulting colouring satisfies (a) and (b), since each $y \in Y$ only sends edges of colour $a \in A$ to one of $U_a(0)$ and $U_a(1)$ (the one corresponding to $g(y)_a$), and similarly for each $x \in X$ and $b \in B$.

Our task is therefore to construct functions f and g such that every edge xy has at least one permitted colour. In this case it is easy to show that such a pair of functions exists.

Lemma 2.4. *For every $d \geqslant 2$, there exist functions*

$$f, g : \{0,1\}^d \to \{0,1\}^d$$

such that for every $x, y \in \{0,1\}^d$, there exists $i \in [d]$ such that either

$$f(x)_i = y_i \qquad \text{or} \qquad g(y)_i = x_i.$$

Proof. Choose the functions f and g uniformly at random, independently of one another. The expected number of pairs (x, y) such that $f(x)_i \neq y_i$ and $g(y)_i \neq x_i$ for every $i \in [d]$ is equal to 1, and if f and g are both the identity function then it is equal to 2^d, so there must exist a choice of f and g for which it is zero, as claimed. $\square$

We have already explained how to deduce Theorem 2.1 from Lemma 2.4, but for completeness let us spell out the details. Let

$$f\colon \{0,1\}^A \to \{0,1\}^B \qquad \text{and} \qquad g\colon \{0,1\}^B \to \{0,1\}^A$$

be the functions given by Lemma 2.4 applied with $d = |A| = |B|$, and observe that every edge xy with $x \in \{0,1\}^A$ and $y \in \{0,1\}^B$ has at least one permitted colour, since either

$$f(x)_b = y_b \text{ for some } b \in B \qquad \text{or} \qquad g(y)_a = x_a \text{ for some } a \in A.$$

We may therefore define a colouring χ of the edges between X and Y simply by choosing a permitted colour for each.

To see that this colouring satisfies property (a), simply note that for each colour $a \in A$ and each vertex $y \in Y$, we have

$$N_a(y) \cap X \subset \big\{ x \in X : x_a = g(y)_a \big\} = U_a(g(y)_a).$$

Property (b) follows by symmetry, and so this completes the proof of Theorem 2.1.

3. The general version of the key lemma

In order to iterate the procedure described in the previous section, we need to be able to start with a k-colouring φ of K_n formed of r-partite graphs, and construct a $(k+t)$-colouring χ of K_N formed of $(r+1)$-partite graphs, for suitable values of the parameters. We will set $d = t/\log k$, choose a collection $\mathcal{F}$ of sets of k colours so that

$$|\mathcal{F}| \geqslant e^{-2t}\binom{k+t}{t} \qquad \text{and} \qquad |S \bigtriangleup T| \geqslant 2d$$

for every $S, T \in \mathcal{F}$ with $S \neq T$, and for each such pair choose d-sets

$$A \subset S \setminus T \qquad \text{and} \qquad B \subset T \setminus S,$$

of colours to use between the corresponding pair of copies X and Y of φ. As in Section 2, the fact that each colour in $A \cup B$ is used in exactly one of X and Y will ensure that there are no monochromatic triangles with vertices in three different copies of φ, and that the edges of each colour in χ form an $(r+1)$-partite graph.

Our aim is therefore to define a colouring between X and Y such that, for each vertex $y \in Y$ and each colour $a \in A$, we have

$$N_a(y) \cap X \subset U_a(i) \quad \text{for some } i \in [r],$$

where $V(H_a) = U_a(1) \cup \cdots \cup U_a(r)$ is a proper r-colouring of the graph H_a formed by the edges in X of colour a, and similarly for each vertex $x \in X$ and colour $b \in B$. To do so, let us map each element $x \in X$ to the hypercube $\{1, \ldots, r\}^A$ by encoding, for each colour $a \in A$, the part $U_a(i)$ containing x, and similarly map Y to $\{1, \ldots, r\}^B$. We will choose functions

$$f \colon \{1, \ldots, r\}^A \to \{1, \ldots, r\}^B \qquad \text{and} \qquad g \colon \{1, \ldots, r\}^B \to \{1, \ldots, r\}^A$$

that encode the set of permitted colours for each edge xy, in such a way that every edge has a permitted colour. This leads us naturally to the following generalisation of Lemma 2.4, which was originally proved (though unfortunately not stated explicitly in this form) by Alon, Ben-Eliezer, Shangguan and Tamo (see [2, Section 4]).

Lemma 3.1. *For every $d \geqslant \lceil 2r \log r \rceil^2$, there exist functions*

$$f, g : \{1, \ldots, r\}^d \to \{1, \ldots, r\}^d$$

such that for every $x, y \in \{1, \ldots, r\}^d$, there exists $i \in [d]$ such that either

$$f(x)_i = y_i \qquad \text{or} \qquad g(y)_i = x_i.$$

We will apply this lemma in almost exactly the same way as in the previous section to define the colouring between X and Y. The rest of this section is devoted to the proof of this lemma; the idea is to take f to be a blow-up of a random colouring on the first $\ell = \lceil 2r \log r \rceil$ coordinates, and then to pick g greedily to give colours to the edges that are missed by f.

Lemma 3.2. *For every $d, \ell, r \in \mathbb{N}$ with $\ell \geqslant 2r \log r$ and $d \geqslant 2\ell r \log r$, there exists a function*

$$h : \{1, \ldots, r\}^\ell \to \{1, \ldots, r\}^d$$

such that for every $y \in \{1, \ldots, r\}^d$ and every set $S \subset \{1, \ldots, r\}^\ell$ with $|S| = \ell$, there exists $z \in S$ and $i \in [d]$ such that

$$h(z)_i = y_i.$$

Proof. We simply choose h randomly, and observe that the expected number of pairs (S, y) for which the claimed property fails to hold is at most

$$r^d \binom{r^\ell}{\ell} \left(1 - \frac{1}{r}\right)^{d\ell} < \exp\left(d \log r + \ell^2 \log r - \frac{d\ell}{r}\right) \leqslant 1,$$

since $\ell \geqslant 2r \log r$ and $d \geqslant 2\ell r \log r$. $\square$

We can now easily deduce Lemma 3.1 by blowing up the colouring given by h to form f, and filling in the gaps greedily using g.

Proof of Lemma 3.1. Let h be the function given by Lemma 3.2, applied with $\ell = \lceil 2r \log r \rceil$, and observe that for each $y \in \{1, \ldots, r\}^d$, the set

$$T(y) = \big\{ z \in \{1, \ldots, r\}^\ell : h(z)_i \neq y_i \text{ for all } i \in [d] \big\}$$

satisfies $|T(y)| < \ell$. Let $T(y) = \{z(1), \ldots, z(m)\} \subset \{1, \ldots, r\}^\ell$, define

$$f(x_1, \ldots, x_d) = h(x_1, \ldots, x_\ell)$$

for every $x = (x_1, \ldots, x_d) \in \{1, \ldots, r\}^d$, and let $g(y) \in \{1, \ldots, r\}^d$ be any vector such that

$$g(y)_i = z(i)_i \qquad \text{for each } i \in [m].$$

To see that these two functions have the desired property, let $x, y \in \{1, \ldots, r\}^d$, set $z = (x_1, \ldots, x_\ell)$, so $f(x) = h(z)$, and observe that if $z \notin T(y)$ then $f(x)_i = y_i$ for some $i \in [d]$. On the other hand, if $z \in T(y)$, then $g(y)_i = z_i = x_i$ for some $i \in [m]$, as required. $\square$

4. The iteration

In this section we will complete the proof of Theorem 1.1. The induction step is as follows.

Lemma 4.1. *Let $n, k, t \in \mathbb{N}$ with $t \geqslant \lceil 2r \log r \rceil^2 \log k$. Suppose there exists a colouring of the edges of K_n with k triangle-free r-partite graphs. Then for some*

$$N \geqslant e^{-2t} \binom{k+t}{t} n,$$

there exists a colouring of the edges of K_N with $k+t$ triangle-free $(r+1)$-partite graphs.

We will take $e^{-2t}\binom{k+t}{t}$ disjoint copies of the colouring φ of K_n, using pairwise far apart sets of k colours for each. We choose this collection of sets of colours greedily, as before.

Lemma 4.2. *If $t \geqslant d \log k$, then there exists a collection $\mathcal{F}$ of subsets of $[k+t]$ of size k with*

$$|\mathcal{F}| \geqslant e^{-2t} \binom{k+t}{t} \qquad \text{and} \qquad |S \triangle T| \geqslant 2d$$

for every $S, T \in \mathcal{F}$ with $S \neq T$.

Proof. Choose the sets in $\mathcal{F}$ greedily, one at a time, from the k-subsets of $[k+t]$ at distance at least $2d$ from all earlier elements of $\mathcal{F}$. We can continue doing so as long as

$$|\mathcal{F}| \sum_{i=0}^{d-1} \binom{t}{i} \binom{k}{i} < \binom{k+t}{t},$$

and since $d \leqslant t / \log k$ it follows that we can continue until $|\mathcal{F}| \geqslant e^{-2t}\binom{k+t}{t}$, as claimed. $\square$

The lemma now follows by the method described in the previous two sections.

Proof of Lemma 4.1. Let $\mathcal{F}$ be given by Lemma 4.2, applied with $d = \lceil 2r \log r \rceil^2$, and for each $S \in \mathcal{F}$, take a disjoint copy of φ, using the colours S, and with vertex set X_S. For each pair $S, T \in \mathcal{F}$, choose sets

$$A \subset S \setminus T \qquad \text{and} \qquad B \subset T \setminus S,$$

with $|A| = |B| = d$, noting that this is possible because $|A \triangle B| \geqslant 2d$. We will use only colours in the set $A \cup B$ between X_S and X_T. It follows that the new edges of each colour form a bipartite graph, and it will therefore suffice to define a colouring between X_S and X_T, using the colours $A \cup B$, that does not create any monochromatic triangles in $X_S \cup X_T$.

To do so, let f and g be the functions given by Lemma 3.1, and map X_S and X_T to $\{1,\dots,r\}^A$ and $\{1,\dots,r\}^B$ in the usual way, using proper r-colourings of the colours in φ. Say that the colour i is permitted for xy if either

$$f(x)_i = y_i \text{ and } i \in B \qquad \text{or} \qquad g(y)_i = x_i \text{ and } i \in A,$$

and observe that every edge has at least one permitted colour, by our choice of f and g. We may now define the colouring χ between X_S and X_T by choosing any permitted colour for each edge. Note that the resulting colouring has the property that

$$N_a(y) \cap X_S \subset \big\{ x \in X_S : x_a = g(y)_a \big\} = U_a\big(g(y)_a\big)$$

for each $y \in X_T$ and $a \in A$, and similarly

$$N_b(x) \cap X_T \subset \big\{ y \in X_T : y_b = f(x)_b \big\} = U_b\big(f(x)_b\big)$$

for each $x \in X_S$ and $b \in B$. Since $U_c(i)$ is an independent set in colour c for every $c \in A \cup B$ and $i \in [r]$, it follows that there are no monochromatic triangles in $X_S \cup X_T$, as required. □

We can now easily deduce Theorem 1.1.

Proof of Theorem 1.1. Set $s = k^{1/3}/\log k$ and $t = k^{2/3}\log k$, and apply Lemma 4.1 for each $r \in [s]$. We obtain a colouring of the edges of K_n with k triangle-free graphs for some

$$n \geqslant e^{-2st} \prod_{r=1}^{s} \binom{rt}{t} \geqslant \left(\frac{s!}{e^{2s}}\right)^t \geqslant \left(\frac{s}{e^3}\right)^{st} = \left(\frac{k^{1/3}}{e^3 \log k}\right)^k,$$

as required. □

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